A HEALTH RECOMMENDATION SYSTEM: FOOD IMAGE RECOGNITION-BASED APPROACH

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Abstract. In contemporary society, a healthy lifestyle is a goal for many, yet our regular diets often include excessive amounts of unhealthy, junk food, potentially harming our health. This study introduces a food AI system capable of identifying everyday food choices and providing nutritional insights and healthier food alternatives to consumers. The first part of the thesis involved the development of a food image recognition system using the Xception model, achieving a 74.84% accuracy rate in the test dataset. Following this, a recommendation system was constructed based on the K-nearest neighbors algorithm. This system inputs a given food recipe and recommends the top eight most similar, yet healthier, food recipes. This research plays a crucial role in promoting healthier eating habits among individuals, steering them towards better health in the future.

Keywords: Health recommendation, Food recognition, K-nearest neighbors algorithm, Xception model, Recommender system

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Table of Contents

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1 Introduction

In an era where the prevalence of lifestyle-related health issues is on the rise, the quest for a healthy lifestyle has never been more pressing. Central to this pursuit is the consumption of nutritious food, a challenge in the face of the convenience and pervasiveness of junk food. This thesis introduces an innovative approach to addressing this challenge: a Health Recommendation System utilizing food image recognition to guide users towards healthier dietary choices.

The core purpose of this research is to develop and assess a system capable of recognizing various foods through image analysis and subsequently providing nutrition-based recommendations. This project aims to bridge the gap between technological advancement and dietary health, harnessing the power of AI to deliver tangible benefits to individuals seeking to improve their eating habits.

The thesis pivots around a crucial research question: Can a food image recognition-based AI system effectively identify food items and suggest healthier alternatives that align with nutritional guidelines? The objectives set to address this question include:

1. Developing an AI model that accurately recognizes food items from images.
2. Creating an algorithm that can recommend healthier food options based on recognized items.
3. Evaluating the system's performance and user satisfaction.

The thesis is structured to facilitate a comprehensive understanding of the system's development and efficacy. Following this introduction, the next section conducts a thorough Literature Review that contextualizes this study within the existing body of research, highlighting the novelty and necessity of the proposed system. Subsequently, an Overview of the Food Recognition System is presented, reviewing the work that was done in part A. We then delve into the Health Recommendation System, detailing problem it aims to solve, the methodologies applied, the implementation process, and the evaluation of results. A Timeline section tracks the project's progression, setting the stage for the Conclusion, which encapsulates the research outcomes, reflects on the system's impact, and offers avenues for future research.

By aligning cutting-edge technology with the pursuit of health, this thesis not only contributes to the academic field but also provides practical value to individuals seeking to improve their dietary choices.

2 Literature Review

The burgeoning interest in the intersection of technology and health has given rise to a diverse body of literature on AI-powered food recognition and recommendation systems. This literature review examines the current state of research in these areas, seeking to understand the depth of existing methodologies, evaluate their effectiveness, and identify opportunities for innovation within this thesis.

Initial research has focused on the technical viability of food image recognition. Kawano and Yanai's (2014) pioneering work on FoodCam, a real-time mobile food recognition system, laid the groundwork for subsequent systems, demonstrating the potential of applying deep learning to food recognition. The evolution of convolutional neural networks (CNNs), particularly the introduction of architectures like Xception, which has been adopted in this research, has significantly advanced the field (Chollet, 2017). Studies by Singh and Lee (2016) have highlighted the efficacy of CNNs in food image classification tasks, with accuracy levels that reinforce the model selection for this thesis.

In terms of dietary recommendations, literature reveals an increasing trend towards personalized nutrition where AI systems are tailored to individual dietary needs (Ferrara et al., 2019). However, many of these systems rely heavily on user input and dietary logs, as seen in the works of Elsweiler et al. (2017), who investigated the link between food logging and health outcomes. This thesis aims to simplify user interaction by directly analysing food images, thereby reducing the reliance on manual logging.

Recent advancements in the field have been marked by the introduction of graph convolutional networks (GCNs) in food recommendation systems. Gao et al. (2021) introduced the Food recommendation with Graph Convolutional Network (FGCN), which goes beyond traditional content-based and collaborative filtering methods. FGCN leverages complex ingredient-ingredient, ingredient-recipe, and recipe-user relations to enhance the accuracy of food recommendations. By modeling high-order connectivity through information propagation mechanisms, FGCN signifies a leap in the personalization of dietary recommendations, integrating deeper relational insights into the recommendation process.

Critically, while the accuracy of food image recognition systems has improved, the integration of these systems with health recommendation components is still in its infancy. A review by Ge, Ricci, and Massimo (2015) on recommender systems highlights this gap, noting that most food-related recommenders do not account for visual dietary information. This gap presents an opportunity for this research to contribute by integrating a visual recognition system with a health-oriented recommendation system.

The methodology chosen for this thesis is informed by the review of both domains. The application of the Xception model for image recognition is predicated on its balance of depth and computational efficiency, making it suitable for mobile deployment as evidenced by performance metrics in the literature (Chollet, 2017). The incorporation of the K-nearest neighbors algorithm for the recommendation system is supported by its successful application in domains where similarity-based recommendations are pertinent (Altman, 1992). Additionally, the insights from the FGCN model present an opportunity to explore the integration of graph-based learning for enriched context in dietary recommendations.

Through this review, the research positions itself at the convergence of food image recognition and health-based food recommendation. It is clear from the literature that while steps have been made towards individual components of such a system, there remains a pronounced need for a unified system that can both accurately identify food items from images and provide health-conscious recommendations. This thesis intends to fill that need, contributing a novel perspective to the existing body of knowledge.

In summary, the literature reviewed not only validates the chosen approach of this research but also underscores the innovation it proposes to introduce. By examining the performance and limitations of existing systems, this thesis aims to address the critical need for seamless integration of AI in dietary management, aligning with the current technological advancements and the growing demand for health-centric solutions.

3 Overview of the Food Recognition System

In Part A of this project, we focused on the development of a Food Recognition System, a pivotal component in the realm of AI-driven dietary management. The central challenge addressed was the accurate identification and analysis of food items from images, a task complicated by the intrinsic diversity in the appearance of food. This aspect of food variability, encompassing differences in shape, colour, texture, and presentation style, necessitated the development of a sophisticated model capable of recognizing and interpreting these subtle yet critical variations.

The approach to developing this system involved a combination of data acquisition, pre-processing, and the employment of advanced machine learning techniques. The Food-101 dataset, comprising 101,000 images across 101 distinct food categories, was selected for its extensive diversity and volume, providing a robust foundation for effective supervised learning. The Xception architecture, a deep learning model pre-trained on ImageNet with 36 convolutional layers, was chosen for its proven efficacy in large-scale image classification tasks. This selection was informed by its balance of depth and computational efficiency, which is crucial for practical application in real-world scenarios.

The implementation phase, as detailed in Part A, was conducted on the Google Colab Pro platform, utilizing the computational capabilities of NVIDIA A100 GPUs. In this phase, Python 3 and TensorFlow were the primary tools, facilitating the import and training of the Xception model. The dataset was strategically split, with 75,750 images allocated for training the model and 25,250 for testing purposes. To enhance the training process, additional images were generated using the ‘ImageDataGenerator’ command, ensuring a more comprehensive learning experience for the model. Notably, the model was trained for whole-image classification, adapting to the dataset's lack of bounding boxes for specific food items.

Several modifications were made to the model to optimize its performance. For instance, the activation functions of the final three layers were switched from ReLU to Sigmoid, a change that proved effective in improving the accuracy of the model. The training process was conducted in two stages: initially for 60 epochs, where the model reached a convergence in accuracy at 67.04%, and subsequently, the pre-trained model was further refined over another 60 epochs, culminating in a final accuracy of 82.02%.

Upon completion of the training, the model's performance was rigorously tested. The system achieved an accuracy of 74.84% on the testing set, a promising result considering the complexity of food image recognition and the absence of more detailed image annotations like bounding boxes. This outcome not only demonstrated the model's capability in identifying a wide array of food items but also highlighted the potential and limitations of current AI technologies in the nuanced field of food recognition.

In conclusion, the development of the Food Recognition System in Part A laid a solid foundation for the broader objective of this thesis. It addressed key challenges in the field of AI-based food analysis and set the stage for integrating this technology with health-oriented applications, as will be explored in this report. The learnings and outcomes from this phase are crucial for understanding the potential and directions for future research in AI-driven dietary management and health promotion.

4 Health *Recommendation System*

4.1 Problem Definition

The core problem that the Health Recommendation System aims to address is the gap in providing accessible, personalized, and health-conscious food recommendations based on an individual's actual dietary intake. In an age where diet-related health issues are increasingly prevalent, there is a pressing need for technological solutions that not only identify what individuals are eating but also guide them towards healthier dietary choices in a personalized and context-aware manner.

The significance of this problem intertwines the realms of health and technology. On the health front, the escalating rates of diet-related diseases such as obesity, diabetes, and heart disease underscore the urgency of facilitating healthier eating habits. Current methods of dietary tracking and recommendation are often manual, time-consuming, and lack personalization, making them less effective for individuals with diverse dietary needs and preferences.

From a technological perspective, the challenge lies in creating a system that is intelligent enough to understand the nuances of various food items from a nutritional standpoint and recommend alternatives that are not only healthier but also align with the user's dietary preferences and constraints. This requires advanced algorithms capable of processing complex nutritional data and user profiles, and delivering recommendations that are both accurate and contextually relevant.

Therefore, the Health Recommendation System is designed to fill this critical void by leveraging sophisticated AI techniques to analyse food items and provide customized healthy eating suggestions. This system stands at the intersection of dietary health and AI-driven personalization, representing a novel approach in the journey towards improved health and well-being through smarter food choices.

4.2 Methodology and Implementation

The methodology for the Health Recommendation System cantered on leveraging a K-nearest neighbors (KNN) algorithm, a renowned technique in machine learning for finding similarity-based matches. The system was designed using Python 3, with the inclusion of NumPy and pandas libraries for efficient data processing and linear algebra operations. Key components from the scikit-learn library, specifically NearestNeighbor and MinMaxScalar, were integral to the model's functioning.

The starting point involved the meticulous preprocessing of the recipes dataset, which comprised 522,517 recipes spanning 312 categories. This dataset offered comprehensive details for each recipe, including cooking times, servings, ingredients, nutrition, instructions, and more. For this system, the focus was narrowed down to nutritional data. Key nutritional attributes, such as Calories, FatContent, SaturatedFatContent, CholesterolContent, SodiumContent, CarbohydrateContent, FiberContent, SugarContent, and ProteinContent, along with RecipeId and RecipeServings, were isolated for subsequent analysis.

The KNN model was trained to identify the 50 recipes most similar to a given recipe based on these nutritional attributes. A user could input a RecipeId, prompting the system to output the 50 closest matches. To enhance the system's utility in promoting healthy eating, a unique healthiness score was developed. This score assigned different weights to each nutritional attribute, reflecting their relative contribution to a recipe's overall healthiness. Positive weights were assigned to desirable nutritional elements like FiberContent and ProteinContent, while other properties received negative weights, aligning with dietary guidelines that recommend their moderation.

The scoring mechanism was carefully calibrated to ensure balance and fairness. The average values for each nutritional attribute across all recipes were computed. These averages were then used to normalize the weights, preventing any single attribute from disproportionately influencing the healthiness score. Notably, a coefficient of 3.5 was applied to the positive attributes to counterbalance the predominance of negative weights, thus ensuring more balanced final scores.

4.3 Results and Evaluation

Upon implementation, the system demonstrated the ability to consistently recommend eight similar and relatively healthy recipes for any given input recipe. The integration of the KNN model with the custom healthiness scoring algorithm resulted in recommendations that were not only similar in terms of ingredients and nutritional profile but also aligned with healthier eating guidelines. These recommendations always scored higher than the input recipe on healthiness.

The system’s performance was promising, but further testing and user feedback are essential for comprehensive evaluation. Future enhancements could include refining the healthiness score algorithm to account for emerging nutritional research or user dietary preferences. Additionally, incorporating user feedback mechanisms could provide valuable insights into the system's real-world applicability and areas for improvement.

Overall, while the initial results are encouraging, ongoing assessment and refinement are crucial for ensuring the system remains robust, user-friendly, and aligned with evolving nutritional science and dietary needs.

5 Timeline

The execution of this project was meticulously structured into two phases, each aligned with an academic term. This approach facilitated focused development on each system component, ensuring thoroughness and quality. Below is the detailed timeline that outlines the major milestones, deadlines, and the actual progress made during each phase.

Term 2: Development of the Food Recognition System

Weeks 1-2: Project Planning and Literature Review

- Activities: Defining project scope, conducting an extensive literature review.
- Progress: Completed as planned, establishing a strong foundation for the project.

Weeks 3-5: Data Acquisition and Pre-processing

- Activities: Sourcing the Food-101 dataset, preprocessing and preparing data for model training.
- Progress: Achieved on schedule, with efficient data handling ensuring a smooth transition to the next phase.

Weeks 6-8: Model Training and Evaluation

- Activities: Training the Xception model, evaluating its accuracy and performance.
- Progress: Minor delays encountered due to computational challenges, but these were swiftly resolved, slightly extending into the ninth week.

Weeks 9-10: Final Report Writing and Project Report for Part A

- Activities: Compiling research findings, drafting, and finalizing the Part A report.
- Progress: Completed, with additional time from the eighth week utilized to enhance the quality of the report.

Term 3: Development of the Health Recommendation System

Weeks 1-2: Planning for the Health Recommendation System

- Activities: Conceptualizing the design and functionality of the recommendation system.
- Progress: Executed as planned, with clear goals set for the system development

Weeks 3-5: System Design and Implementation

- Activities: Developing the KNN-based recommendation system, integrating the healthiness score algorithm.
- Progress: On track, with successful integration of the components.

Weeks 6-8: Testing, Refinement, and Evaluation

- Activities: Conducting system tests, refining based on evaluation, and iterating improvements.
- Progress: Extended into the early part of the ninth week due to the iterative nature of testing and refinement.

Weeks 9-10: Final Report Writing and Project Report for Part B

- Activities: Documenting the development process, results, and evaluations of the Health Recommendation System.
- Progress: Completed, with extra time from the eighth week aiding in thorough documentation and review.

This timeline reflects a realistic and structured approach to the project, accommodating unforeseen challenges and ensuring that each phase of development received the necessary attention and rigor. The minor deviations encountered were effectively managed, demonstrating adaptability and commitment to maintaining the project's high standards.

6 Conclusion

This thesis has described the development of a two-part system: a Food Recognition System and a Health Recommendation System. The key findings from this research underscore the significant potential of integrating artificial intelligence with nutritional guidance.

The Food Recognition System, developed using the Xception model, demonstrated an accuracy of 74.84% in identifying food items from images. This achievement highlights the system's capability to effectively process and recognize diverse food images, a task complicated by the inherent variability in food appearance. The Health Recommendation System, leveraging a K-nearest neighbors algorithm, adeptly recommended healthier recipe alternatives based on nutritional content, effectively combining technological innovation with health consciousness.

The contribution of this work to the field is multifaceted. It not only advances the domain of AI in food recognition but also bridges a critical gap by integrating these capabilities with health-oriented recommendations. This dual approach is particularly relevant in today's health-conscious society, providing a practical tool for individuals seeking to make informed dietary choices.

The journey of this research was marked by both learning and challenges. The complexities involved in accurately classifying a wide range of food items, compounded by the nuances of nutritional data interpretation, presented significant hurdles.

Based on the findings of this research, several recommendations are proposed:

- Utilizing bounding boxes: It is anticipated that the system's accuracy would significantly improve with the incorporation of bounding boxes in image labeling and during the model training process.
- User-centric features: Incorporating user feedback and preferences into the Health Recommendation System could personalize the recommendations, making them more relevant and practical.
- Integrating the Food Recognition and Health Recommendation Systems: By methodically aligning the Food Recognition System with the Health Recommendation System, we can create a streamlined process where a food image inputted into the Food Recognition System is used to generate a label. This label then serves as a direct input for the Health Recommendation System, ultimately yielding tailored recommendations for healthier food options.

In conclusion, this research represents a significant step forward in the fusion of AI with dietary management, offering a promising tool for promoting healthier eating habits. It sets a foundation for future exploration and innovation in this vital field, with the potential to contribute substantially to public health and well-being.

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